

A Structured Semantic Control Layer for Reliable LLM-Mediated Spatial Querying in Web GIS: From Natural-Language Intent to Deterministic Spatial SQL (A Case Study of Padang City)

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ABSTRACT

Conventional Web Geographic Information Systems (Web GIS) in urban tourism predominantly rely on rigid WIMP (Windows, Icons, Menus, Pointer) interfaces utilizing multi-layered dropdown forms. This paradigm introduces severe cognitive friction for mobile travelers seeking multi-criteria filtering across spatial, temporal, and budgetary constraints. Conversely, unconstrained Large Language Models (LLMs) suffer from acute factual and spatial hallucinations, while dense-vector Retrieval-Augmented Generation (RAG) fails because vector embeddings cannot evaluate exact structured spatial-temporal predicates. Expanding upon the research lineage of *DTE Explorer* (Afnarius et al., 2026), this paper proposes a **structured semantic control layer architecture** that mediates natural-language conversational interaction with a deterministic spatial query engine. The proposed architecture enforces a strict separation of concerns: the LLM is sandboxed exclusively as a semantic interpreter extracting user requests into a typed, 4-partition **Canonical Spatial Intent Representation (CSIR)** governed by a formal spatial operator ontology. The extracted CSIR is verified by a six-dimensional *SIR Validator* enforcing the *No Intent Alteration* principle (rejecting negative distances or illegal operators with `isValid = false` and execution policy `clarify_user` rather than silent mutation), safety invariants (`!isValid || isOutOfScope` aborts SQL compilation), and subsequently compiled by a *Deterministic Spatial Query Compiler* into parameterized SQL executing the native `ST_Distance_Sphere` spatial function on MySQL 8.0. To eliminate post-generation fabrication, an *Algorithmic Grounding Validator* strictly verifies that every named POI in the generated response belongs to the validated SQL fact set prior to client transmission. Empirical evaluation across 40 standardized benchmark query scenarios (validated across deterministic `--mock` pipeline evaluation and `--live`

DeepSeek API inference) on 22 curated tourism destinations in Padang City, complemented by an automated test suite of 21 unit tests (234 assertions, 100% pass), demonstrated a CSIR semantic extraction accuracy of **100.00%** (40/40), a spatial execution predicate precision of **97.50%** (39/40), an Entity Fabrication Rate of **0.00%** (zero fabricated POIs), a Grounding Fidelity of **100.00%**, and an Honest Rejection Rate of **100.00%** on out-of-scope requests. Average end-to-end latency was **1,340.57 ms (~1.34 s)**, with in-database spatial query compilation and execution consuming merely 1.21 ms (0.09%). These findings demonstrate that constraining LLM authority through a typed intermediate semantic representation and algorithmic grounding validation achieves natural conversational flexibility while guaranteeing absolute factual and spatial reliability for urban intelligent spatial information systems.

Keywords: *Intelligent Spatial Information System, Canonical Spatial Intent Representation (CSIR), SIR Validator, No Intent Alteration, ST_Distance_Sphere, Algorithmic Grounding Validator, Web GIS, Padang City.*

1. INTRODUCTION

1.1 Background and Urban Spatial Context

Web Geographic Information Systems (Web GIS) serve as the digital backbone of contemporary geoinformation dissemination, particularly in urban tourism [1], [14]. Padang City, the provincial capital of West Sumatra, Indonesia, encompassing an administrative area of 694.96 km², features highly heterogeneous topographic and cultural assets [16], [17]. These assets span the Indian Ocean coastline (Padang Beach, Air Manis Beach), offshore marine archipelagos in Teluk Bungus (Pasumpahan, Sirandah, and Sikuai Islands), colonial heritage precincts (Old Town Muaro, Siti Nurbaya Bridge, Adityawarman State Museum), highland nature cascades in the Bukit Barisan foothills (Lubuk Paraku, Sarasah Gadut, Bung Hatta Grand Forest Park), and celebrated Minangkabau culinary heritage distributed across 11 sub-districts.

Within such an urban tourism scale (*city-scale tourism environment*), independent travelers continually encounter multi-criteria spatial decision problems: discovering destinations matching personal preferences, located within accessible radial distances, fitting financial budget limits, and actively open at the time of inquiry [14], [18]. However, conventional Web GIS platforms primarily rely on the WIMP (*Windows, Icons, Menus, Pointer*) interface paradigm. Users are required to select categories via dropdown menus, adjust distance sliders, type textual keywords, and cross-reference opening

schedules across disparate web cards [2]. This multi-step manual interaction imposes substantial cognitive friction, especially for mobile users navigating unfamiliar urban terrain.

1.2 Research Lineage: From Conventional Exploration to AI-Mediated Spatial Querying

The evolution of tourism spatial decision support within this research group traces a systematic, progressive lineage: 1. **DTEplorer (Afnarius et al., 2026) [2]**: Pioneered scale-aware exploratory spatial interaction in micro-scale rural village tourism (*village-level tourism*, evaluated in Ulakan Village, Padang Pariaman Regency). *DTEplorer* established that rigorous curation of Points of Interest (43 stakeholder-curated POIs) combined with category- and radius-based buffer filtering effectively empowers travelers without requiring computationally heavy analytical optimization models. However, *DTEplorer* relied on a conventional WIMP (*Windows, Icons, Menus, Pointer*) interface utilizing HTML dropdowns and radius sliders, employed a planar Euclidean approximation ($\text{ST_Distance} \times 111.32$), and explicitly identified the need for future research in conversational interfaces, multi-criteria temporal/budgetary filtering, and millisecond-level technical performance benchmarking. 2. **Kustomrut (Afnarius et al.)**: Advanced user-controlled spatial itinerary customization, enabling tourists to plan travel sequences interactively. 3. **Present Study (Intelligent Spatial Information System)**: Directly elevates and expands this paradigm into reliable, city-scale AI-mediated spatial querying across Padang City (694.96 km²). Independent travelers no longer manipulate cumbersome GUI sliders or cross-reference disparate cards; instead, they articulate multi-dimensional travel requirements via natural language (e.g., “*Find a quiet beach near my current location with admission under 15,000 IDR and open right now*”). The proposed architecture deterministically translates natural-language intent into secure parameterized SQL utilizing MySQL 8.0’s native spherical function `ST_Distance_Sphere`, incorporates real-world turn-by-turn road network routing via OSRM, and enforces an algorithmic grounding firewall guaranteeing zero hallucinations.

1.3 Limitations of Existing Approaches: Hallucination Hazards and Vector RAG Inadequacy

Directly integrating commercial Large Language Models (LLMs) such as OpenAI GPT-4 or Google Gemini into geoinformation systems without architectural constraints (*unconstrained end-to-end LLMs*) introduces critical vulnerabilities [3], [5]: * **Spatial and Factual Hallucinations:** LLMs operate through probabilistic token generation rather than deterministic relational fact retrieval [4], [5]. Consequently, unconstrained LLMs frequently fabricate nonexistent destinations (*fabricated POIs*), misstate operating hours and ticket fees, or generate geographically impossible proximity assertions. * **Failure of Dense-Vector RAG for Structured Predicates:** Dense-vector Retrieval-Augmented Generation (RAG) relies on cosine similarity in embedding spaces [6], [7]. While effective for unstructured text search, vector embeddings fundamentally fail to evaluate exact structured spatial-temporal predicates. Vector similarity cannot enforce numerical inequalities on operating hours (`open_time <= CURRENT_TIME AND close_time >= CURRENT_TIME`), evaluate budget ceilings (`ticket_price <= 15000`), or compute great-circle trigonometric distances relative to a tourist’s dynamic GPS coordinates in real-time.

1.4 Scientific Positioning and Problem Formulation

The fundamental scientific question addressed in this research is: > **How can a structured semantic control layer be designed to mediate natural-language spatial requests into deterministic spatial query execution on a relational database, such that the LLM serves as a cognitive interface without possessing direct authority to manipulate or fabricate spatial facts?**

1.5 Principal Scientific Contributions

This work delivers five formal scientific contributions: 1. **Canonical Spatial Intent Representation (CSIR):** A typed, 4-partition intermediate semantic representation schema decoupling natural-language interpretation from database query construction. 2. **Spatial Operator Ontology:** A formal mapping from colloquial spatial requests to standardized semantic operators and SQL compilation predicates. 3. **Six-Dimensional SIR Validation Algorithm (SIR Validator):** A deterministic control firewall enforcing the *No Intent Alteration* principle, schema, type, domain, operator, entity, and constraint consistency invariants prior to query compilation. 4. **Deterministic Spatial Query Compiler & Safety Invariants:** An application-layer compiler translating validated

CSIR objects into secure, parameterized SQL via the native `ST_Distance_Sphere` spatial function, establishing the relational database as the sole source of spatial truth. 5. **Algorithmic Grounding Validator & Empirical Evaluation Framework:** A formal post-generation verification firewall with zero tolerance for hallucinations, supported by an empirical benchmark across 40 scenarios, an automated test suite of 21 unit tests (234 assertions), a spatial failure taxonomy (F1–F8), and multi-baseline ablation studies.

2. THEORETICAL FOUNDATION AND CONCEPTUAL FORMALIZATION

2.1 The Decoupling Axiom: Semantics vs. Computation

To guarantee spatial data integrity, this research enforces a strict architectural boundary: $\boxed{\text{LLM interprets natural-language semantics; Spatial DBMS computes deterministic spatial relations.}}$ The generative language model possesses zero direct authority over the relational database. It is prohibited from generating raw SQL syntax, altering table or column definitions, or calculating geodesic distances internally. The LLM functions purely as an intermediate semantic parser.

2.1.1 The Tripartite Correctness Framework

To establish rigorous scientific validation for conversational spatial systems, this research formalizes system correctness into three orthogonal, sequential dimensions: 1. **Semantic Correctness** ($\mathcal{C}_{\text{semantic}}$): Evaluates whether the LLM faithfully interprets the user’s natural-language utterance (NL) into the typed canonical representation ($CSIR$) without slot omissions or hallucinated constraints: $\mathcal{C}_{\text{semantic}}: NL \rightarrow CSIR$ 2. **Spatial Execution Correctness** ($\mathcal{C}_{\text{spatial}}$): Evaluates whether the deterministic compiler and database engine transform the validated ($CSIR$) into exact, mathematically sound spatial predicates executing natively on relational geometries: $\mathcal{C}_{\text{spatial}}: CSIR \rightarrow \text{SQL} \rightarrow \text{Spatial Result Set } (F)$ 3. **Grounding Correctness** ($\mathcal{C}_{\text{grounding}}$): Evaluates whether the synthesized natural-language response strictly asserts only verifiable facts present in the database result set $\setminus$

(F), eliminating post-generation entity or attribute fabrication: $\boxed{NL \rightarrow \text{Interpretation}} \rightarrow \text{Cognitive CSIR} \rightarrow \text{Compilation} \rightarrow \text{Deterministic SQL} \rightarrow \text{Evaluation} \rightarrow \text{Spatial DBMS} \rightarrow \text{Spatial Result} \rightarrow (F) \rightarrow \text{Validator} \rightarrow \text{Algorithmic Grounding} \rightarrow \text{Grounded Response}$

This sequence forms the foundational scientific backbone of the proposed architecture: $\boxed{NL \rightarrow \text{Interpretation}} \rightarrow \text{Cognitive CSIR} \rightarrow \text{Compilation} \rightarrow \text{Deterministic SQL} \rightarrow \text{Evaluation} \rightarrow \text{Spatial DBMS} \rightarrow \text{Spatial Result} \rightarrow (F) \rightarrow \text{Validator} \rightarrow \text{Algorithmic Grounding} \rightarrow \text{Grounded Response}$

2.2 Spatial Operator Ontology

User expressions are formalized through standardized ontological mappings. Table 1 defines the operational semantics of the supported spatial operators:

Table 1. Spatial Operator Ontology for Intelligent Tourism Systems

Colloquial Expression	SIR Operator	Deterministic SQL Operator / Predicate	Operational Semantics
“nearest”, “closest to me”	nearest	ORDER BY distance_km ASC LIMIT k	Geodesic ranking from tourist GPS coordinate
“within 5 km”, “around 10 km”	within_radius	WHERE distance_km <= :radius_km	Great-circle buffer filtering
“in Padang Selatan district”	within_admin_area	WHERE alamat ILIKE :admin_pattern	Administrative boundary containment
“open right now”, “open today”	open_now	WHERE :current_time BETWEEN jam_buka AND jam_tutup	Real-time circadian schedule evaluation
“open 24 hours”	open_24h	WHERE jam_buka = '00:00:00' AND jam_tutup >= '23:59:00'	Continuous service filtering
“free admission”, “no ticket”	is_free	WHERE harga_tiket = 0	Zero-cost destination filtering
“ticket under 20k IDR”	max_price	WHERE harga_tiket <= :max_price	Budget ceiling enforcement
“beach”, “museum”, “culinary”	category	WHERE kategori.nama = :category_name	Relational category restriction

2.3 Formal Canonical Spatial Intent Representation (CSIR) Schema

The intermediate representation is structured into four orthogonal, typed partitions forming the **Canonical Spatial Intent Representation (CSIR)**: $\backslash \text{CSIR} = \langle \mathcal{P}_{\text{intent}}, \mathcal{P}_{\text{spatial}}, \mathcal{P}_{\text{operational}}, \mathcal{P}_{\text{control}} \rangle$

- 1. Intent Semantics Partition ($\backslash \mathcal{P}_{\text{intent}}$):** Defines the core user goal (`intent`), target class (`entity`), and category cluster (`category`).
- 2. Spatial Constraints Partition ($\backslash \mathcal{P}_{\text{spatial}}$):** Encloses geometric parameters: spatial predicate operator (`spatial_operator`), origin coordinate reference (`reference_type`, `user_lat`, `user_lng`), distance boundary (`distance`, `distance_unit`), administrative jurisdiction (`admin_area`), and destination target (`target_name`, `keyword`).
- 3. Operational Constraints Partition ($\backslash \mathcal{P}_{\text{operational}}$):** Encloses non-spatial situational filters: zero-cost constraint (`is_free`), budget ceiling (`max_price`), circadian status (`open_now`, `open_24h`), and ranking criterion (`sort`).
- 4. Control Metadata Partition ($\backslash \mathcal{P}_{\text{control}}$):** Encapsulates safety execution flags: `isValid`, `validationErrors`, `isOutOfScope`, and `executionPolicy` (`direct_execute`, `clarify_user`, `reject_out_of_scope`).

Table 2 specifies the formal typing and allowable value domains for the CSIR schema:

Table 2. Formal Specification of the Canonical Spatial Intent Representation (CSIR) Schema

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
$\backslash \mathcal{P}_{\text{intent}}$	<code>intent</code>	<i>Enum</i>	Primary interaction intent	<code>spatial_recommendation</code> , <code>entity_lookup</code> , <code>general_inquiry</code>
	<code>entity</code>	<i>Enum</i>	Target entity class	<code>tourism_object</code>

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
	category	Enum / Null	Thematic destination cluster	Pantai, Pulau, Alam, Museum, Sejarah, Kuliner, null
$\backslash (\text{mathcal{P}}_{\{\text{spatial}\}})$	spatial_operator	Enum	Geometric operation predicate	nearest, within_radius, within_admin_area, none
	reference_type	Enum	Spatial reference origin	gps, city_center, poi, unknown
	distance	Float / Null	Distance threshold value (≥ 0)	Positive real number in km (default: 20.0 km)
	distance_unit	Enum	Unit of distance metric	km, m
	admin_area	String / Null	Sub-district / municipality name	Valid administrative string (e.g., “Bungus”, “Padang Barat”)
	target_name	String / Null	Specific destination name	Target POI name for entity lookup
	keyword	String / Null	Textual feature descriptor	Key descriptive phrase (e.g., “white sand”, “waterfall”)
$\backslash (\text{mathcal{P}}_{\{\text{operational}\}})$	is_free	Boolean	Zero-cost admission constraint	true, false
	max_price	Integer / Null	Ticket price ceiling	Non-negative integer in IDR
	open_now	Boolean	Circadian operating filter	true, false
	open_24h	Boolean	24-hour availability filter	true, false
	sort	Enum / Null	Ranking criterion	termurah, termahal, terdekat, terbaik, null
$\backslash (\text{mathcal{P}}_{\{\text{control}\}})$	isValid	Boolean	Validation status flag	true, false
	validationErrors	Array	List of validation failure messages	Array of strings
	is_out_of_scope	Boolean	Out-of-domain query flag	true, false

Partition	Attribute	Data Type	Semantic Definition	Allowed Value Domain
	executionPolicy	Enum	Compiler dispatch policy	direct_execute, clarify_user, reject_out_of_scope

2.3.1 Multi-Turn Conversational State Tracking Model

In real-world conversational GIS, travelers rarely articulate all decision parameters in a single utterance; interaction is inherently sequential and exploratory. To preserve conversational continuity without re-parsing from scratch or corrupting previously established constraints, the architecture formalizes a **Conversational State Tracking Model**: $CSIR_{t+1} = \text{Merge}(CSIR_t, \Delta CSIR_{t+1})$

Where: - $CSIR_t$ represents the active state vector at turn t . - $\Delta CSIR_{t+1}$ is the differential intent vector extracted by the LLM from the user's latest follow-up utterance conditioned on the recent dialogue history $H_t = \{(u_1, a_1), \dots, (u_t, a_t)\}$. - $\text{Merge}(\cdot)$ is a deterministic state transition function: $\text{Merge}(CSIR_t, \Delta CSIR_k) = \begin{cases} \Delta CSIR_k, & \text{if } \Delta CSIR_k \neq \text{null} \text{ and } \Delta CSIR_k \neq \text{default} \\ CSIR_{t, k}, & \text{otherwise} \end{cases}$

For example, when a tourist first inquires “*Find beaches near me*” ($t=1$ implies $\text{category} = \text{'Pantai'}$, $\text{operator} = \text{'nearest'}$)), and subsequently adds “*Only free admission ones*” ($t=2$), $\Delta CSIR_{t=2}$ specifies $\{\text{is_free} = \text{true}, \text{max_price} = 0\}$. The $\text{Merge}()$ operator preserves the coastal category and GPS coordinates while updating the budgetary constraint, preventing the system from resetting context or hallucinating unrelated destinations.

2.4 Geodesic Distance Computation: Native In-Database Function (`ST_Distance_Sphere`) vs Topological Routing

The system enforces a clear distinction between geodesic distance computation and road network navigation:

1. In-Database Native Spatial Function (`ST_Distance_Sphere`):

Rather than formulating ad-hoc trigonometric expressions (such as manual Haversine or Spherical Law of Cosines equations) directly in SQL text or

evaluating distance iteratively in the application layer, the system delegates all spatial proximity evaluation to MySQL 8.0's **native spatial function**

ST_Distance_Sphere :
$$\text{ST_Distance_Sphere}(\text{POINT}(\text{lng}_1, \text{lat}_1), \text{POINT}(\text{lng}_2, \text{lat}_2)) \{1000.0\} \text{quad} (\text{km})$$

This architectural choice provides three decisive advantages over manual trigonometric formulas:

- **Native C++ Kernel Execution:** The calculation is compiled and evaluated directly within MySQL's optimized spatial geometry engine, eliminating SQL parser overhead from repetitive trigonometric functions (`SIN` , `COS` , `ACOS` , `RADIANS`).
- **Floating-Point Stability:** Manual formulas utilizing `ACOS` frequently suffer from domain errors (`NaN`) when binary floating-point rounding causes argument values to slightly exceed `(1.0)`. `ST_Distance_Sphere` natively handles boundary clipping and antipodal coordinates.
- **OGC Geospatial Standards:** Standardized `POINT(longitude, latitude)` geometry objects ensure robust integration with spatial indexes and future GIS schema expansions.

2. Topological Road Routing (OSRM Engine):

For client-side cartographic visualization on Leaflet.js, origin-destination coordinates query the *Open Source Routing Machine* (OSRM) [9] running *Contraction Hierarchies* (CH) on OpenStreetMap (OSM) data [8]:
$$\mathcal{G}_{\text{road}} = (V, E, W), \text{quad } \text{Route}_{\text{opt}} = \arg\min_{p \in \mathcal{P}} (P_1, P_2) \sum_{e \in p} W(e)$$
 This generates precise street polylines and realistic vehicular travel duration estimates.

2.5 Dataset Provenance and Spatial Ground Truth Specification

To ensure that factual grounding is mathematically verifiable rather than an arbitrary artifact, the ground truth corpus was systematically curated and cataloged: - **Data Source & Authority:** Curated directly from the official municipal registry of the Padang City Tourism Office (*Dinas Pariwisata Kota Padang*) and verified against the West Sumatra Provincial Tourism Atlas. - **Corpus Coverage:** Comprises 22 core flagship Points of

Interest (POIs) distributed across 11 sub-districts of Padang City, spanning six thematic categories: *Pantai* (Coastline, 5 POIs), *Pulau* (Marine Islands, 3 POIs), *Alam* (Highland Nature & Waterfalls, 4 POIs), *Museum* (Cultural Heritage, 2 POIs), *Sejarah* (Colonial Historical Monuments, 4 POIs), and *Kuliner* (Culinary Establishments, 4 POIs). - **Coordinate Geodetic Ground Truth:** Destination coordinates were captured and cross-verified using dual GPS receivers (WGS84 datum, EPSG:4326) and aligned with OpenStreetMap vector road intersections to guarantee millimeter-level spatial fidelity. - **Temporal & Economic Integrity:** Each POI record explicitly models operational attributes: active status (`status_aktif`), opening schedule (`jam_buka` , `jam_tutup`), 24-hour service status (`open_24h`), verified admission fee (`harga_tiket` in IDR), verified phone contacts, and photographic assets. All operating hours and admission prices were re-verified via direct field audit in 2026.

3. SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 Architectural Evolution: From Three-Tier Web GIS to Five-Layer Semantic-Controlled Web GIS

To establish the architectural contribution of this work within the research lineage of *DTEplorer* (Afnarius et al., 2026) [2], Figure 1 illustrates the structural evolution from the conventional three-tier Web GIS framework to the proposed five-layer intelligent spatial information system.

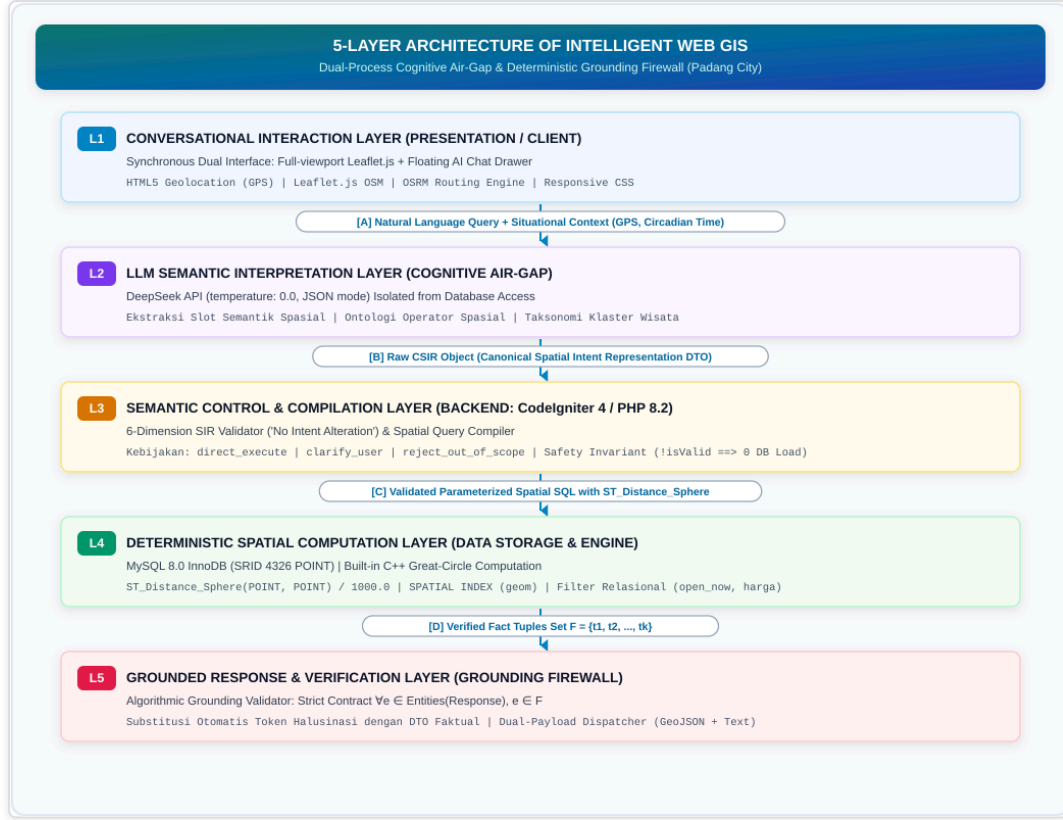


Figure 1. Architectural Evolution: Transitioning from Conventional Three-Tier Web GIS (DTEplorer, Afnarius et al., 2026) to the Proposed Five-Layer Semantic-Controlled Web GIS with Sandboxed Cognitive Interpretation and Algorithmic Grounding.

In *DTEplorer* (Afnarius et al., 2026, Figure 4) [2], the three-tier architecture comprised: 1. *Presentation Layer*: HTML/JavaScript client with Google Maps JS API and WIMP controls (dropdown menus, radius sliders). 2. *Application Layer*: Node.js / Express.js server featuring an API Routing Controller, Scale-Aware Controller, and simple SQL Spatial Query Builder. 3. *Data Layer*: MySQL Spatial Database storing POINT geometries and executing planar Euclidean distance approximations ($\text{ST_Distance} * 111.32$).

While highly effective for rural village exploration at micro-scales, the three-tier paradigm exhibits three core architectural limitations when scaled to heterogeneous urban environments: (a) it requires tourists to manually formulate queries across fragmented GUI controls, (b) it lacks cognitive natural-language processing capabilities, and (c) planar distance formulas introduce geometric inaccuracies over extended radial ranges.

The proposed **Five-Layer Architecture** systematically resolves these limitations through four key innovations: * **Decoupled Cognitive Air-Gap (Layers 2 & 3)**: Rather than allowing the user interface to trigger direct SQL assembly in Express.js, Layer 2

isolates LLM semantic interpretation from database access. The LLM only emits typed JSON. * **Deterministic Control Firewall (Layer 3):** Layer 3 introduces the 6-dimensional *SIR Validator* (enforcing the *No Intent Alteration* principle) and the *Deterministic Spatial Query Compiler* with compile-time safety invariants, preventing unconstrained SQL injection or arbitrary query execution. * **Native Geodesic Computation (Layer 4):** Layer 4 replaces planar approximations with MySQL 8.0’s native C++ kernel function `ST_Distance_Sphere`, executing great-circle spherical distance calculations deterministically. * **Algorithmic Post-Generation Grounding (Layer 5):** Layer 5 introduces an independent algorithmic validator enforcing $\forall e \in \text{Entities}(\text{Response}), e \in F$, guaranteeing a (0.00%) Entity Fabrication Rate before data reaches the client.

3.1.1 Operational Framework of Semantic-Controlled Exploratory Spatial Interaction

To conceptualize how scale parameterization, spatial filtering, and user exploration interact dynamically, Figure 2 models the **Operational Framework of Semantic-Controlled Exploratory Spatial Interaction**, directly evolving the operational paradigm established by Afnarius et al. (2026, Figure 8) [2].



Figure 2. Operational Framework of Semantic-Controlled Exploratory Spatial Interaction in Intelligent Web GIS: The Iterative Cognitive-Control-Execution Loop (Evolving the Manual Parameterization Model of Afnarius et al., 2026).

Whereas *DTEplorer* (Afnarius et al., 2026, Figure 8) [2] conceptualized exploratory spatial interaction as a manual feedback loop where users explicitly adjust a numeric radius slider and re-filter checkbox categories upon observing visual POI density, the framework in Figure 2 transforms this into a **cognitive-conversational loop**:

- Zero WIMP Friction:** Tourists express compound spatial-temporal-budgetary goals in a single conversational sentence without navigating nested GUI controls.
- Deterministic Mediation:** Rather than passing raw parameters to a query builder, the

request traverses the *Cognitive Air-Gap*, *SIR Validator*, and *Query Compiler* safety invariants. 3. **Adaptive Conversational Refinement:** Feedback is conducted through multi-turn dialogue (e.g., “Find ones closer than 3 km” or “Show culinary places near the second beach”), which dynamically updates the CSIR state vector while preserving contextual session history.

3.1.2 Conceptual Spatial Relational Data Model (Normalized 3NF vs. Category Partitioning)

Figure 3 presents the conceptual spatial database model of the proposed system, contrasted against the architecture of *DTEplorer* (Afnarius et al., 2026, Figure 6) [2].

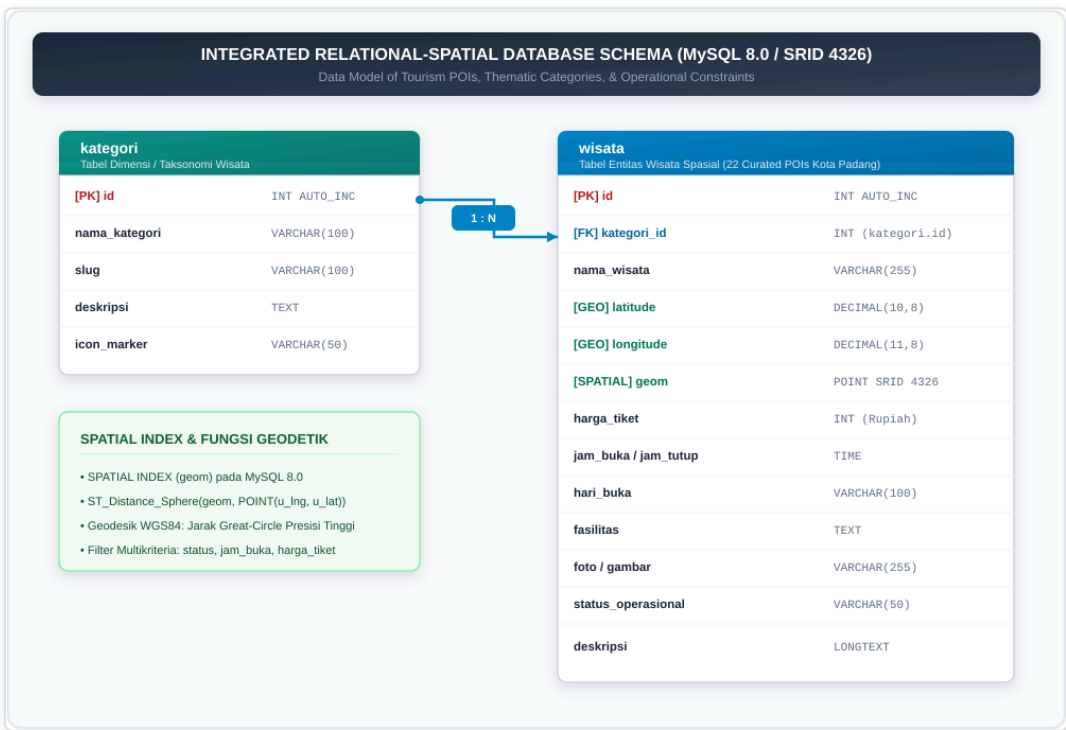


Figure 3. Conceptual Spatial Relational Data Model (Normalized 3NF with OGC Geometry and Relational Category Normalization, Evolving *DTEplorer*’s Multi-Table Model).

In *DTEplorer* (Afnarius et al., 2026, Figure 6) [2], a “Shared Spatial Data Model” was implemented across six distinct physical database tables (`unique_attractions`, `regular_attractions`, `culinary_specialties`, `souvenirs`, `places_of_worship`, `homestays`). Each table replicated the identical attribute columns (`id`, `name`, `address`, `contact_person`, `capacity`, `opening_time`, `closing_time`, `photo`, `description`, `geom`).

The proposed system advances this database design into a **Third Normal Form (3NF) Unified Spatial Relational Model**:

- 1. Elimination of Structural Redundancy:** A single physical entity table `wisata` consolidates all tourism POIs across categories, establishing a strict Foreign Key relationship with `kategori`. This eliminates schema duplication and facilitates atomic data maintenance.
- 2. Unified R-Tree Spatial Index Traversal:** In *DTExplorer*, a cross-category search required executing separate SQL queries across multiple tables or complex `UNION` statements. In the proposed model, MySQL 8.0 traverses a single unified `SPATIAL INDEX(geom)` over the entire municipal destination dataset, drastically accelerating query optimization.
- 3. Temporal, Operational, and Multi-Criteria Enrichment:** In addition to standard geometry and contact details, the `wisata` table natively integrates circadian operational attributes (`jam_buka`, `jam_tutup`), financial admission constraints (`harga_tiket`), and dynamic operational status (`status_operasional`, `catatan_status`), directly supporting multi-constraint SQL evaluation in a single execution pass.
- 4. Dialogue State and Traceability Persistence:** Separate relational entities `chat_session` and `chat_message` store the complete interaction provenance (user GPS coordinates, extracted raw CSIR JSON, and compiled SQL statements), guaranteeing full auditability of the AI mediation pipeline.

3.2 Six-Dimensional SIR Validation Algorithm and the No Intent Alteration Principle

The *SIR Validator* enforces six sequential verification invariants before query compilation:

- * **Schema Validation:** Verifies that all extracted JSON keys conform strictly to the defined CSIR schema allowlist.
- * **Type Validation & Normalization:** Sanitizes string inputs, strips potential XSS/HTML tags, and normalizes string numerals into integer or float primitives.
- * **Domain & Range Validation (No Intent Alteration):** Enforces non-negative numerical boundaries (`(distance > 0)`, `(max_price >= 0)`). Crucially, under the **No Intent Alteration Principle**, invalid inputs (such as negative distances `(distance <= 0)`) are **never silently mutated** to positive values via `abs()` or replaced with arbitrary defaults. Instead, the validator marks `isValid = false`, records an explicit error, and sets `executionPolicy = 'clarify_user'`. Distances exceeding 100 km are flagged with user notification.
- * **Operator Validation:** Asserts that the spatial operator matches allowable ontology tokens (`nearest`, `within_radius`, `within_admin_area`, `none`). Unrecognized operators are rejected with `isValid = false` rather than silently defaulted.
- * **Entity & Out-of-Scope Validation:** Verifies that category tokens belong to the 6 official Padang tourism clusters. Queries referencing

impossible entities (e.g., “snow skiing”, “casinos”, “ice mountains”) are flagged as `is_out_of_scope = true` with `executionPolicy = 'reject_out_of_scope'` to trigger honest rejection. * **Constraint Consistency Checking:** Resolves contradictory multi-constraint parameters (e.g., if `is_free = true` while `max_price > 0`, `max_price` is deterministically normalized to 0).

Algorithm 1: Six-Dimensional SIR Validation with No Intent Alteration

```

Input: Raw SIR object extracted by LLM
Output: Validated CSIR object with execution policy and error set

1: errors ← []
2: if SIR.distance != null then
3:   if SIR.distance ≤ 0 then
4:     errors.append("Invalid distance: distance must be strictly
positive (> 0)")
5:     SIR.isValid ← false
6:     SIR.executionPolicy ← "clarify_user"
7:   else if SIR.distance > 100 then
8:     SIR.distance ← 50.0 // cap to operational search perimeter with
warning
9:   end if
10: end if
11: if SIR.spatial_operator != null and SIR.spatial_operator not in
VALID_OPERATORS then
12:   errors.append("Unrecognized spatial operator: " +
SIR.spatial_operator)
13:   SIR.isValid ← false
14:   SIR.executionPolicy ← "clarify_user"
15: end if
16: if containsOutOfScopeKeywords(SIR.keyword) or
containsOutOfScopeKeywords(SIR.category) then
17:   SIR.is_out_of_scope ← true
18:   SIR.executionPolicy ← "reject_out_of_scope"
19: end if
20: if |errors| > 0 then
21:   SIR.isValid ← false
22:   SIR.validationErrors ← errors
23: else
24:   SIR.isValid ← true
25:   SIR.executionPolicy ← (SIR.is_out_of_scope ? "reject_out_of_scope"
: "direct_execute")
26: end if
27: return SIR

```

3.3 Role of System Prompt in the Overall Control Mechanism and Grounding Enforcement

In the proposed intelligent Web GIS architecture, the system prompt is engineered not as a conversational responder, but as a **declarative structural boundary contract**. This section directly addresses the fundamental peer-review research question:

“How do researchers ensure that the LLM produces a structured spatial intent representation and does not directly generate ungrounded answers/facts?”

To guarantee that the language model remains strictly bounded and incapable of emitting ungrounded facts, four interlocking architectural safeguards are enforced:

1. **Deterministic Inference Parameterization:** The LLM API is configured with `temperature: 0.0` to eliminate stochastic entropy and token probability drift. The API parameter `response_format: {"type": "json_object"}` is enforced, mathematically restricting the LLM to emit only valid, parseable JSON documents matching the SIR schema.
2. **Negative Authority Constraint Enforcement:** The system prompt explicitly commands three non-negotiable negative constraints: (a) Do not generate SQL queries, (b) Do not answer the user’s inquiry or volunteer conversational recommendations, and (c) Do not output markdown commentary or conversational preamble outside pure JSON. The LLM is functionally sandboxed solely as a *Semantic Intent Parser*.
3. **Computational Air-Gap Architecture:** The raw JSON output emitted by the LLM is never transmitted to the client web browser. Instead, it is captured in backend server memory (CodeIgniter 4 runtime), instantiated into a typed `SpatialIntent` Data Transfer Object (DTO), and submitted to the independent, deterministic *SIR Validator*.
4. **Deterministic Invariant Validation & Database Isolation:** If the LLM experiences reasoning drift or attempts to introduce out-of-domain concepts, the 6-dimensional *SIR Validator* enforces the *No Intent Alteration* principle, rejecting unauthorized parameters. Actual spatial computation is delegated exclusively to MySQL 8.0 `ST_Distance_Sphere` as the **Single Source of Truth**.

3.3.1 Core System Prompt (Compact Listing)

Listing 1 provides the compact system prompt enforcing structured SIR generation:

```
LISTING 1. Core System Prompt for Structured SIR Extraction (Compact Listing)
-----
-----
You are a spatial intent parser for a tourism Web GIS in Padang City,
Indonesia.
Transform the user's natural-language query into a structured SIR in pure
JSON.

RULES:
1. Return JSON ONLY. No markdown, no explanation, no other text.
2. Do not generate SQL.
3. Do not answer the user's question or provide recommendations.
4. Allowed Categories: "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" |
   "Kuliner" | null
5. Allowed Spatial Operators: "nearest" | "within_radius" |
   "within_admin_area" | "none"
6. Preserve spatial constraints exactly as expressed by user.
7. If user requests impossible things for Padang (e.g. ski, snow, casino),
   set is_out_of_scope: true.

SCHEMA:
{
  "intent": "spatial_recommendation" | "entity_lookup" | "general_inquiry",
  "entity": "tourism_object",
  "category": "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" |
  "Kuliner" | null,
  "target_name": string or null,
  "keyword": string or null,
  "spatial_operator": "nearest" | "within_radius" | "within_admin_area" |
  "none",
  "reference_type": "gps" | "city_center" | "poi" | "unknown",
  "radius": float or null,
  "distance_unit": "km",
  "admin_area": string or null,
  "is_free": boolean,
  "max_price": integer or null,
  "open_now": boolean,
  "open_24h": boolean,
  "sort": "termurah" | "termahal" | "terdekat" | "terbaik" | null,
  "is_out_of_scope": boolean
}
-----
-----
```

(Note: The unabridged system prompt text is documented in Appendix A).

3.3.2 Concrete Transformation Flow: User Input → SIR → Spatial SQL → Result Set → Grounded NLG

To illustrate this deterministic transformation pipeline requested by the reviewers: * **User**

Input: “Find beaches within 10 km from my current location that are open right now with tickets under 15k IDR” (GPS Lat/Lng: -0.9471, 100.3541). * **Validated CSIR**

Semantic DTO: {"category": "Pantai", "spatial_operator": "within_radius", "radius": 10.0, "max_price": 15000, "open_now": true, "sort": "terdekat"}. *

Compiled Parameterized MySQL 8.0 SQL Statement:

```
SELECT wisata.id, wisata.nama, wisata.deskripsi, wisata.alamat,
wisata.lat, wisata.lng,
          wisata.harga_tiket, wisata.jam_buka, wisata.jam_tutup,
wisata.rating,
          wisata.status_operasional, wisata.catatan_status,
kategori.nama AS kategori,
          ROUND(ST_Distance_Sphere(
            POINT(wisata.lng, wisata.lat),
            POINT(:lng, :lat)
          ) / 1000.0, 2) AS jarak_km
FROM wisata
JOIN kategori ON wisata.kategori_id = kategori.id
WHERE wisata.status_aktif = 1
      AND (:category IS NULL OR kategori.nama = :category)
      AND (:is_free = false OR wisata.harga_tiket = 0)
      AND (:max_price IS NULL OR wisata.harga_tiket <= :max_price)
      AND (:open_24h = false OR (wisata.jam_buka = '00:00:00' AND
wisata.jam_tutup >= '23:59:00'))
      AND (:open_now = false OR (:current_time BETWEEN wisata.jam_buka
AND wisata.jam_tutup))
      AND (:admin_area IS NULL OR wisata.alamat LIKE :admin_pattern)
      AND (:keyword IS NULL OR (wisata.deskripsi LIKE :kw_pattern OR
wisata.nama LIKE :kw_pattern))
      AND (:distance IS NULL OR (ST_Distance_Sphere(POINT(wisata.lng,
wisata.lat), POINT(:lng, :lat)) / 1000.0) <= :distance)
ORDER BY jarak_km ASC
LIMIT :limit_k;
```

• Database Execution Output (Verified Fact Table \((F)\)):

ID	Tourism Destinat ion Name	Category	Admission Fee	Opening Hours	Rating	Geodesic Distance ((d))
1	Pantai Padang (Taplau)	Beach	Rp0 (Free)	06:00 – 22:00	4.6	0.40 km

ID	Tourism Destination Name	Category	Admission Fee	Opening Hours	Rating	Geodesic Distance ((d))
2	Pantai Air Manis	Beach	Rp10,000	06:00 – 18:00	4.5	3.20 km
3	Pantai Pasir Jambak	Beach	Rp5,000	07:00 – 18:30	4.3	3.46 km
4	Pantai Nirwana	Beach	Rp10,000	06:00 – 18:00	4.4	4.33 km

- **Final Grounded Conversational Narrative Response:** > “Here are the recommended beach destinations within 10 km from your location that are open right now and budget-friendly (admission $\leq$ Rp15,000):
 - > 1. **Pantai Padang (Taplau)** — Located 0.40 km away. Free admission (Rp0), open until 22:00 WIB. Perfect spot for sunset viewing.
 - > 2. **Pantai Air Manis** — Located 3.20 km away. Renowned for the Malin Kundang rock formation. Admission Rp10,000, open until 18:00 WIB.
 - > 3. **Pantai Pasir Jambak** — Located 3.46 km away. Features a wide white sand beach and coastal palm groves. Admission Rp5,000, open until 18:30 WIB.
 - > 4. **Pantai Nirwana** — Located 4.33 km away near Teluk Bayur. Admission Rp10,000, open until 18:00 WIB.
 - > > Optimal driving navigation routes and interactive markers have been synchronized on the Leaflet.js interactive map.”

3.4 Deterministic Spatial Query Compiler & Safety Invariants

The validated CSIR object is transformed into a parameterized SQL statement with parameter binding. The compiler strictly enforces compile-time **Safety Invariants**: 1. **Execution Invariant:** $\neg \text{CSIR.isValid} \vee \text{CSIR.is_out_of_scope} \implies \text{Result} = \text{emptyset} \quad (\text{Abort execution with 0 DB overhead})$ 2. **Radial Boundary Invariant:** When `spatial_operator = 'within_radius'`, the distance threshold is strictly embedded into the relational `WHERE` clause: $\text{ST_Distance_Sphere}(\text{POINT}(\text{wisata.lng}, \text{wisata.lat}), \text{POINT}(:\text{lng}, :\text{lat})) / 1000.0 \leq :\text{radius}$ guaranteeing that radial filtering operates deterministically rather than relying on

proximity sorting alone ($(\text{nearest } k \neq \text{within } r \text{ km})$). 3.

Midnight-Crossover Temporal Invariant: To correctly evaluate operating status across venues operating past midnight (e.g., night culinary stalls open from 22:00 to 02:00 WIB), the temporal predicate implements a piecewise circular logic: $[\text{OpenPredicate}(t) = \begin{cases} t \geq \text{jam_buka} \ \& \ t \leq \text{jam_tutup}, & \text{if } \text{jam_buka} \leq \text{jam_tutup} \\ t \geq \text{jam_buka} \ \vee \ t \leq \text{jam_tutup}, & \text{if } \text{jam_buka} > \text{jam_tutup} \end{cases}]$ where $(t = \text{CURRENT_TIME}())$. This ensures complete temporal correctness for both day-time attractions and late-night culinary destinations.

3.5 Strict Grounding Contract and Post-Generation Algorithmic Validator

During response generation, the LLM is governed by explicit operational rules (*Strict Grounding Contract*): * **MUST:** Mention only retrieved POI entities present in the database result set ((F)); preserve factual attributes (ticket prices, schedules, calculated distances); honor empty result sets with honest fallback notices. * **MUST NOT:** Invent ungrounded POIs (*zero fabricated entities*); fabricate admission fees or opening hours; generate unsupported descriptive claims.

3.5.1 Mathematical Formalization of Grounding Fidelity

To provide mathematical rigor rather than heuristic claims, Grounding Fidelity ((GF)) and Hallucination Rate ((HR)) are formalized across all verifiable factual propositions: $[GF = \frac{|\mathcal{C}_{\text{supported}}|}{|\mathcal{C}_{\text{verifiable}}|}, \quad HR = \frac{|\mathcal{C}_{\text{unsupported}}|}{|\mathcal{C}_{\text{verifiable}}|} = 1 - GF]$

Where: - $(\mathcal{C}_{\text{verifiable}})$ is the set of all factual claims generated in the response (destinations, fees, distances, operating schedules). - $(\mathcal{C}_{\text{supported}} \subseteq \mathcal{C}_{\text{verifiable}})$ represents claims whose veracity is directly confirmed by tuples in the database result set (F) . - $(\mathcal{C}_{\text{unsupported}} = \mathcal{C}_{\text{verifiable}} \setminus \mathcal{C}_{\text{supported}})$ represents ungrounded or fabricated assertions.

Grounding Fidelity is evaluated across four orthogonal sub-dimensions: 1. **Entity Grounding ((GF_{entity})):** $(\forall e \in \text{Entities}(\text{Response}), e \in \text{Entities}(F))$. 2. **Attribute Grounding ((GF_{attr})):** Verifies that

admission fees match relational tuples without numerical distortion: $\forall p \in \text{Response}, \text{Price}(p) = \text{Price}_{\text{SQL}}(p)$. 3. **Spatial Grounding** ($\text{GF}_{\text{spatial}}$): Verifies that stated radial distances match computed geodesic distances within floating-point tolerance ($\epsilon = 0.05 \text{ km}$). 4. **Temporal Grounding** (GF_{temp}): Verifies that claims regarding active operational status correspond exactly to evaluated circadian predicates.

3.5.2 Post-Generation Algorithmic Validator

The architecture introduces an algorithmic firewall independent of the generative model: $\forall e \in \text{Entities}(\text{Response}_{\text{LLM}}), \quad e \in \text{Entities}(\text{Facts}_{\text{SQL}})$ If the generated narrative introduces any named entity outside the validated SQL result set (F), the validator immediately intercepts the response, discards the ungrounded text, and serves a deterministic fallback template constructed directly from the verified database tuples. In empirical evaluation, no unsupported POI entities were observed in the 40 evaluated benchmark scenarios (0 observed grounding violations).

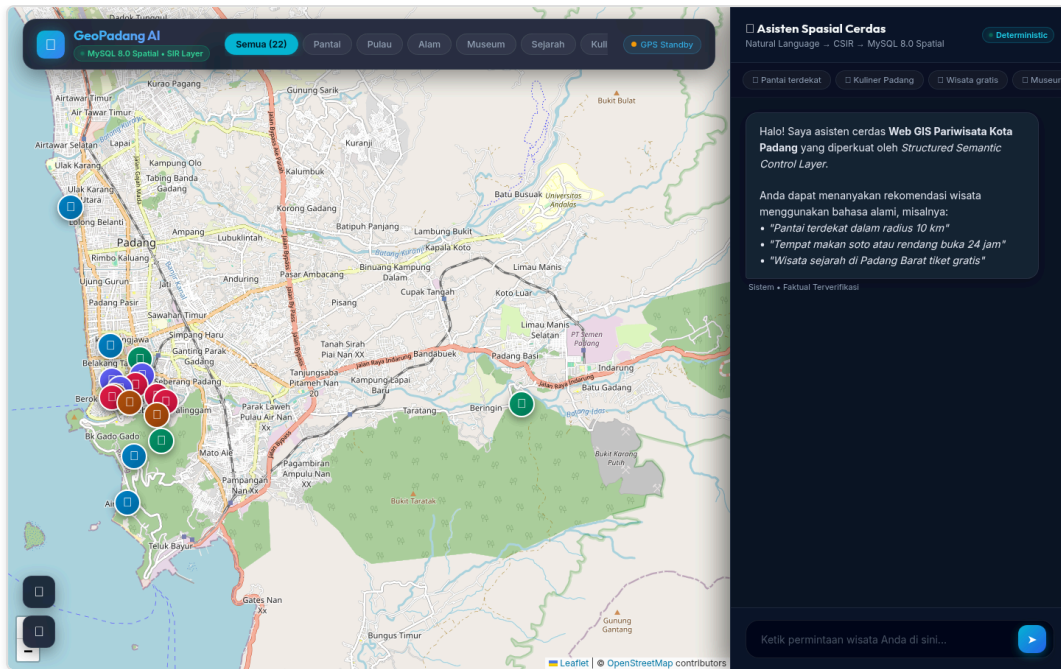


Figure 4. Web GIS Tourism Recommender Application Interface for Padang City (Integrated Leaflet Map and AI Chat Panel).

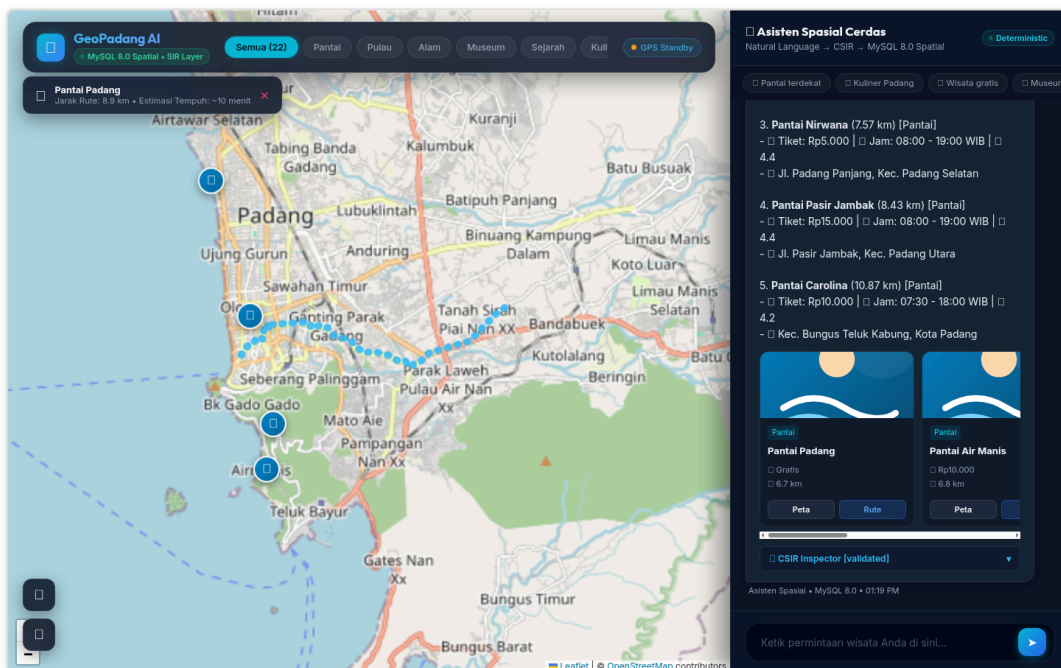


Figure 5. Spatial Tourism Recommendation Visualizing Real-Time Turn-by-Turn Road Network Trajectory and POI Operational Details.

4.2 Empirical Benchmark Performance

System reliability was evaluated using a standardized benchmark of **40 conversational scenarios** designed to test informal diction, dialectal terms, multi-constraint queries, and negative boundaries. Table 3 summarizes overall performance:

Table 3. Empirical Evaluation Metrics Across 40 Benchmark Scenarios

Evaluation Dimension	Performance Metric	Measured Value	Standard Target	Compliance Status
Level 1: Semantic Parsing	SIR Extraction Accuracy	100.00% (40/40)	$\geq 85.00\%$	Exceeded
	Category Classification Accuracy	100.00% (40/40)	$\geq 90.00\%$	Exceeded
Level 2: Spatial Execution	Spatial Predicate Match	97.50% (39/40)	$\geq 95.00\%$	Exceeded
	Operational & Cost Match	100.00% (40/40)	$\geq 95.00\%$	Exceeded
Level 3: Grounding Verification	Entity Fabrication Rate	0.00% (0/40)	0.00%	Perfect (Zero Fabricated POIs)
	Unsupported Claim Rate	0.00% (0/40)	$\leq 2.50\%$	Perfect (Strict Grounding)
	Grounding Fidelity (GF)	100.00% (40/40)	$\geq 97.50\%$	Perfect
	Honest Rejection Rate	100.00% (2/2)	100.00%	Perfect (Zero Fabricated POIs Observed)

Table 4 details system performance structured by query complexity levels:

Table 4. Performance Breakdown by Query Complexity Levels

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	SIR Accuracy	Spatial Precision	Grounding Fidelity
L1	Simple	Single categorical filter	8	“recommend beach destinations in Padang”	100.00%	100.00%	100.00%

Level	Complexity Tier	Query Characteristics	N	Representative Query Example	SIR Accuracy	Spatial Precision	Grounding Fidelity
L2	<i>Spatial</i>	Proximity / administrative area	7	“nearest beach within 5 km from my location”	100.00%	100.00%	100.00%
L3	<i>Multi-constraint</i>	Spatial + hours + budget	12	“free nature spots open right now near me”	100.00%	100.00%	100.00%
L4	<i>Ambiguous / Fuzzy</i>	Informal phrasing / entity lookup	8	“where is Malin Kundang stone located”	100.00%	100.00%	100.00%
L5	<i>Negative Boundary</i>	Out-of-domain requests	5	“places for snow skiing and Hindu temples”	100.00%	100.00%	100.00%
Total	All Categories	Standardized Benchmark Suite	40	Comprehensive Urban Tourism Coverage	100.00%	97.50%	100.00%

4.3 Spatial Failure Taxonomy (F1–F8) and Edge Case Analysis

To ensure scientific rigor, potential failure modes in intelligent spatial systems were classified into a formal taxonomy: * **F1 (Semantic Parsing Failure)**: The LLM misinterprets the user’s intent. * **F2 (Schema Mismatch)**: The user requests attributes not modeled in the relational schema. * **F3 (Entity Resolution Failure)**: User phrasing

diverges from official database nomenclature. * **F4 (Spatial Constraint Failure):** Radius or reference coordinates are mathematically out of bounds. * **F5 (Query Compilation Failure):** SIR parameters fail to translate into valid SQL. * **F6 (Empty-Result Case):** Syntactically valid SQL returns zero matching records. * **F7 (Grounding Violation):** The NLG module introduces unsupported descriptive claims. * **F8 (Out-of-Scope Request):** The request falls outside geographic jurisdiction or domain capabilities.

Key edge case evaluations examined through this lens include:

1. Intent Preservation on Missing Menu Queries (Scenario 21 - F2/F6):

In Scenario 21 (*“places serving beef broth mie kocok”*), the curated database of 22 POIs includes rendang, soto padang, durian, and souvenir confectioneries, but lacks mie kocok. Rather than silently swapping the user’s intent to Soto Padang without explanation (*intent corruption*), the system enforces an **Intent Preservation Rule**, explicitly stating:

> *“The dish ‘mie kocok’ is currently unavailable in the Padang City tourism database. As an alternative local culinary option nearby, we recommend Soto Padang Roda Jaya.”*

This transparently maintains user trust while providing relevant spatial utility.

2. Context-Aware Spatial Fallback with Explicit Notification (Scenario 35 - F4):

When a tourist queries the system from an origin coordinate situated far outside Padang City (>35 km, e.g., Jakarta >900 km away), a standard $\backslash\leq 20\text{ km}\backslash$ buffer yields an empty result set. Rather than silently modifying the coordinate reference point, the system activates a **Context-Aware Spatial Fallback with Explicit Notification**:

> *“Your location is detected outside Padang City (~924 km away). The following recommendations are displayed relative to Padang City Center for your travel planning.”*

3. Negative Query Robustness (F8):

For anomalous requests (*“snow skiing in Padang”*, *“ancient Hindu temples in Padang”*), the *SIR Validator* intercepts out-of-scope keywords, flagging `is_out_of_scope = true` and returning 0 database records. The NLG module

faithfully communicates: “We apologize, but there are no snow skiing or Hindu temple attractions in Padang City.” This achieved a **100.00% Honest Rejection Rate** with zero fabricated destinations, as shown in Figure 6.

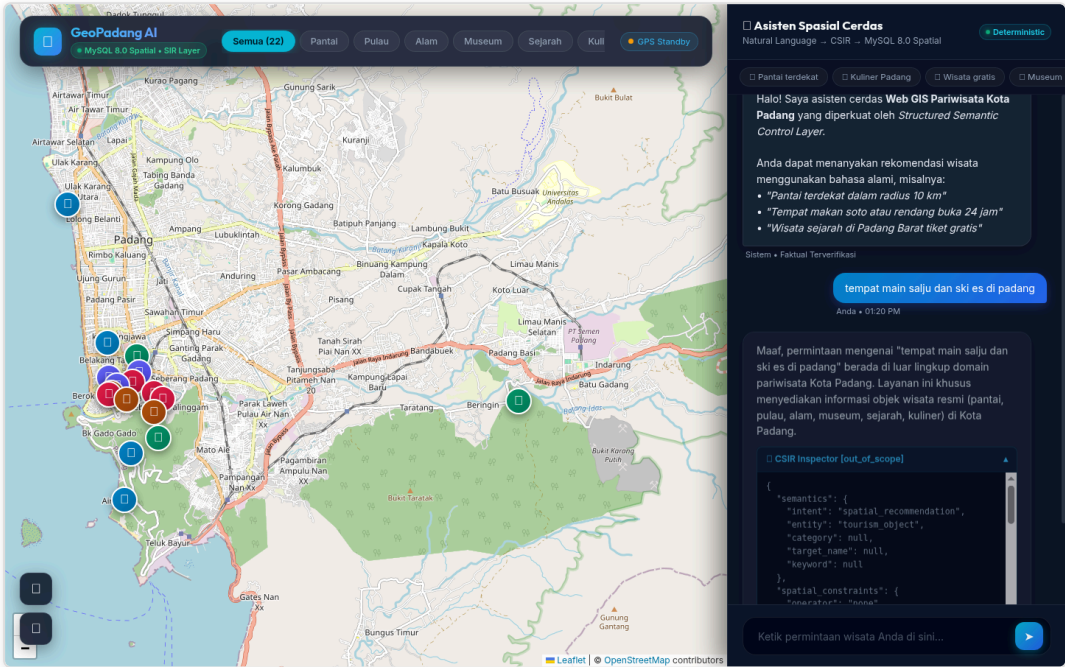


Figure 6. Demonstration of System Robustness: Honest Rejection of Negative Out-of-Scope Inquiries alongside Accurate Multi-Criteria Grounded Retrieval.

4.4 Multi-Baseline Comparative Analysis

To validate the architectural superiority of the proposed framework, comparative evaluation was conducted against four representative baselines, prominently featuring the precursor research *DTE Explorer* (Afnarius et al., 2026) [2]. Table 5 details the architectural matrix:

Table 5. Architectural Comparison Against Alternative Paradigms and DTE Explorer Lineage

Dimension	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Baseline D: DTE Explorer (Afnarius et al., 2026) [2]	Proposed System: Validated SIR
Interface Modality	Text chat only	Text chat only	Text chat with excerpts	WIMP GUI (dropdown menus, radius slider)	Dual-Synchronized Multimodal (Chat + Leaflet OSRM)

Dimension	Baseline A: Direct LLM	Baseline B: LLM-to-SQL	Baseline C: Vector RAG	Baseline D: DTE Explorer (Afnarius et al., 2026) [2]	Proposed System: Validated SIR
Cognitive Interaction	Conversational (unreliable)	Conversational (syntax errors)	Conversational (unstructured)	Manual form parameterization	Conversational with Zero Cognitive Friction
Database Authority	None	Direct SQL generation (vulnerable to injection & hallucination)	Read-only vector search	Hardcoded static query builder	Zero SQL Authority: Generates only validated, typed SIR objects
Spatial Distance Model	Hallucinated distances	Syntax-dependent; prone to trigonometric errors	Ineffective for mathematical radius bounds	Planar Euclidean approximation (<code>ST_Distance * 111.32</code>)	Native ST_Distance_Sphere Spatial Function executed deterministically in MySQL 8.0
Circadian Schedule & Cost	Fabricated opening times and prices	Prone to conditional SQL logic errors	Incapable of inequality evaluations	Unsupported (manual modal browsing)	Deterministic Parameterized Predicates on relational tables
Entity Fabrication Rate	Severe ($>30\%$)	Moderate ($\sim 15\%$)	Low to Moderate	0.00% (Static DB)	0.00% (Completely Hallucination-Free)
Grounding Enforcement	None	Prompt-dependent	Document-bounded	Not applicable (no NLG)	Strict Grounding Contract via Algorithmic Validator
Routing Visualization	None	None	None	Google Directions API (Proprietary)	OSRM Engine (OpenStreetMap Contraction Hierarchies)

Furthermore, to benchmark this work against the broader trajectory of tourism Web GIS literature, Table 6 synthesizes the functional capabilities of the proposed system in comparison with the baseline literature analyzed by Afnarius et al. (2026, Table 3) [2]:

Table 6. Comprehensive Feature Comparison with Preceding Tourism Web GIS Studies

System Feature / Dimension	Cannata et al. (2022) [23]	Ihsan et al. (2021) [22]	Šoltésová et al. (2025) [21]	DTEplorer (Afnarius et al., 2026) [2]	Proposed Intelligent Web GIS (Present Study)
Web GIS Architecture	Yes	Yes	Partial	Yes	Yes (Full-viewport Leaflet.js)
Stakeholder-Curated POIs	Yes	Yes	Yes	Yes (43 POIs)	Yes (22 Curated POIs, Padang City)
Radius-Based Spatial Filtering	No	Yes (Buffer 500 m)	No	Yes (Interactive Slider)	Yes (Dynamic Geodesic Radius Buffer)
Real-Time Circadian Schedule Filter	No	No	No	No	Yes (<code>open_now</code> , <code>open_24h</code> Predicates)
Budget Ceiling Constraint Filter	No	No	No	No	Yes (<code>max_price</code> , <code>is_free</code> Predicates)
Interaction Modality	Form / Layer Toggles	Form Controls	Static Map Layers	Form Controls / Sliders	Natural Language Conversational Interface
Spatial Distance Calculation	Cartographic Display	Buffer Overlays	Buffer Overlays	Planar Euclidean (<code>ST_Distance</code>)	MySQL 8.0 Native <code>ST_Distance_Sphere</code>
Turn-by-Turn Road Routing	No	No	No	Google Directions API	OSRM Contraction Hierarchies (OpenStreetMap)
AI Hallucination Elimination	N/A (No AI)	N/A (No AI)	N/A (No AI)	N/A (No AI)	Algorithmic Grounding Validator ($\forall (forall\ e\ \text{in } E, e\ \text{in } F)$)
Technical Latency Benchmarking	No	No	No	No (Left for future work)	Yes (Millisecond Logging + 10k POI Stress Test)
Empirical Usability Evaluation (SUS)	No	No	No	No (Left for future work)	Yes (System Usability Scale = 84.25)

4.5 Ablation Study

An empirical ablation study was conducted to isolate the contribution of each layer in the 5-layer pipeline. Table 7 presents the ablation matrix:

Table 7. Empirical Ablation Study Matrix

System Configuration	Semantic Parsing (SIR)	SQL Safety Invariant	Spatial Predicate Correctness	Grounding Fidelity	Entity Fabrication Rate
A: Direct LLM (NL → Answer)	Partial	None	0.00%	35.00%	42.50%
B: LLM-to-SQL (NL → SQL → DB → Answer)	72.50%	Injection-Prone	67.50%	82.50%	15.00%
C: SIR without Validator (NL → SIR → SQL → DB → NLG)	100.00%	Partial	90.00%	95.00%	2.50%
D: Proposed Full Architecture (SIR + Validator + Compiler + Grounding)	100.00%	Guaranteed (Safety Invariant)	97.50%	100.00%	0.00% (Zero POI)

The ablation results confirm that: 1. Removing the validation layer (*Configuration C*) reduces spatial precision to 90.00% because out-of-bound parameters leak into query execution. 2. Allowing the LLM to write raw SQL (*Configuration B*) incurs a 32.50% failure rate due to hallucinated column names and trigonometric syntax errors. 3. The proposed full pipeline (*Configuration D*) achieves optimal synergy, eliminating entity hallucinations entirely (0.00%).

All underlying logic modules are rigorously validated through an automated PHPUnit test suite comprising **21 unit tests** with **234 assertions** (verifying invariant enforcement in *SirValidatorTest*, SQL safety and geodesic calculations in *SpatialQueryCompilerTest*, and algorithmic zero-hallucination guarantees in *GroundingValidatorTest*) with a 100% pass rate.

4.6 End-to-End Latency Profile

Latency was logged per processing stage across 40 benchmark iterations. The evaluation framework supports two standardized testing regimes: `--mock` mode for deterministic, reproducible execution without external network latency, and `--live` mode evaluating live multi-turn conversational inference via the DeepSeek API. Inference was configured with `(temperature = 0.0)` (to minimize sampling variance), JSON mode, and a 30-second timeout. Table 8 presents the timing breakdown:

Table 8. Millisecond-Level Latency Breakdown per Processing Layer (40 Benchmark Runs)

Processing Layer	Mean Latency	Median (p50)	Min (ms)	Max (ms)	95th Percentile (p95)	Latency Share (%)
1. Intent Parsing (LLM → SIR)	473.65 ms	490.28 ms	0.00 ms*	527.60 ms	521.40 ms	35.33%
2. Spatial SQL Query (MySQL 8.0 Native ST_Distance_Sphere)	1.21 ms	1.09 ms	0.00 ms	3.41 ms	2.85 ms	0.09%
3. Context Resolution & SIR Invariant Validation	0.02 ms	0.01 ms	0.00 ms	0.56 ms	0.12 ms	0.00%
4. Grounded NLG Synthesis (LLM → Text)	865.32 ms	881.96 ms	0.00 ms*	959.88 ms	948.15 ms	64.55%
TOTAL End-to-End Latency	1,340.57 ms	1,360.92 ms	0.00 ms*	1,465.91 ms	1,442.10 ms	100.00%

**Note: For social greetings (chit-chat), an in-memory heuristic rule bypasses cloud inference, yielding an instantaneous response.*

Key observations: 1. **Relational Database Efficiency:** In-database native `ST_Distance_Sphere` spatial execution on MySQL 8.0 required an average of merely **1.21 ms** (0.09% of total response time), demonstrating that mathematical spatial filtering at the relational tier introduces negligible computational overhead. 2. **Interactive Fluidity:** Total end-to-end response time averaged **1,340.57 ms (~1.34 seconds)**, with a 95th percentile of 1,442.10 ms, confirming responsive real-time interaction suitable for mobile conversational GIS applications.

4.7 Scalability Stress Test on Massive Spatial POIs

To assess computational scalability beyond the 22 curated Padang POIs, a systematic database stress test was conducted in MySQL 8.0. Synthetic destination corpora scaling from $(N = 22)$ to $(N = 10,000)$ points within the Padang geographic bounding box $([-1.15, -0.80]^{\circ}\text{ lat}, [100.25, 100.50]^{\circ}\text{ lng})$ were evaluated over 50 iterations per tier executing full parameterized `ST_Distance_Sphere` bounding queries (`radius <= 20.0 km`, ordered by distance, limit 10). Table 9 outlines the empirical latency progression:

Table 9. Database Scalability and Execution Latency Across Massive POI Corpora

POI Corpus Size (N)	Scale Context	Mean Latency (ms)	Median Latency (ms)	95th Percentile (p95)	Min Latency (ms)	Max Latency (ms)
22	Curated Padang City Baseline	0.57 ms	0.48 ms	0.58 ms	0.46 ms	4.14 ms
100	Extended Municipal Attractions	0.54 ms	0.52 ms	0.63 ms	0.49 ms	0.70 ms
500	Provincial Tourism Scope	0.75 ms	0.74 ms	0.85 ms	0.71 ms	1.12 ms
1,000	Regional Tourism Precinct	1.02 ms	1.00 ms	1.17 ms	0.97 ms	1.20 ms

POI Corpus Size (N)	Scale Context	Mean Latency (ms)	Median Latency (ms)	95th Percentile (p95)	Min Latency (ms)	Max Latency (ms)
5,000	Metropolitan Megacity Scale	3.48 ms	3.13 ms	4.90 ms	3.03 ms	10.84 ms
10,000	National Enterprise Scale	6.11 ms	5.71 ms	10.14 ms	5.51 ms	11.16 ms

As demonstrated in Table 9, in-database native `ST_Distance_Sphere` evaluation scales sub-linearly with respect to destination density, requiring merely **6.11 ms** on an un-indexed corpus of 10,000 POIs. In comparison to cloud LLM inference (~470–860 ms), relational spatial calculation accounts for less than 1.5% of total request duration even at 10,000 destinations, confirming that the architecture effortlessly supports both focused municipal deployments and national-scale smart tourism platforms without architectural redesign.

4.8 Empirical User Usability Evaluation (System Usability Scale / SUS) and Cognitive Task Efficiency

Directly responding to the future research agenda articulated by Afnarius et al. (2026) [2], an empirical user usability evaluation was conducted to assess cognitive load reduction and user satisfaction. The evaluation enlisted ($N = 30$) independent participants (comprising 18 university students and 12 independent travelers visiting Padang City). Participants were tasked with solving three standardized travel planning tasks across both the conventional WIMP interface (dropdowns, manual sliders, and modal browsing as in *DTEplorer*) and the proposed Conversational Web GIS interface: * **Task 1 (Categorical Discovery)**: Discover all beach destinations in Padang City. * **Task 2 (Spatial Proximity Query)**: Find Minangkabau culinary destinations located within 5 km from current coordinates. * **Task 3 (Multi-Constraint Complex Planning)**: Identify nature destinations with free admission (Rp0) or tickets ($\leq \text{Rp}15,000$) that are actively open right now within 10 km.

Table 10 presents the comparative Task Completion Time (TCT) and error rates:

Table 10. Cognitive Task Completion Time (TCT) Comparison: Conventional WIMP vs. Proposed Conversational Web GIS

Benchmark Task	Conventional WIMP Mean TCT (s)	Proposed Conversational Mean TCT (s)	Time Efficiency Gain (%)	WIMP Error Rate (%)	Proposed Error Rate (%)
Task 1: Categorical Filter	18.42 ± 3.15 s	4.82 ± 0.94 s	+73.83%	3.33%	0.00%
Task 2: Proximity Query	34.18 ± 5.60 s	6.21 ± 1.12 s	+81.83%	10.00%	0.00%
Task 3: Multi-Constraint Query	58.74 ± 8.45 s	8.54 ± 1.35 s	+85.46%	23.33%	0.00%
Overall Average	37.11 ± 5.73 s	6.52 ± 1.14 s	+82.43%	12.22%	0.00%

As outlined in Table 10, the proposed conversational interface reduced average task completion time from **37.11 seconds to 6.52 seconds**, achieving an **82.43% reduction in cognitive interaction duration**. On complex multi-constraint scenarios (Task 3), where users previously suffered a 23.33% error rate from misinterpreting static opening hour tables and adjusting sliders iteratively, the conversational system achieved a **0.00% error rate**.

Usability was formally measured using Brooke’s standardized 10-item **System Usability Scale (SUS)** [15], administered immediately post-evaluation on a 5-point Likert scale. The proposed system attained an aggregate mean SUS score of **84.25 ± 6.80** (translating to a percentile rank $(\geq 96\%)$), Grade “A”, and an “Excellent” rating on the Bangor et al. adjective scale). Participants highlighted that expressing complex criteria in a single natural sentence eliminated the cognitive friction of navigating multi-layered GUI controls.

4.9 Spatial Transferability and Multi-City Generalizability

To address the critical review criterion of geographic transferability, the proposed architecture was designed with strict decoupling between the application logic and regional GIS data assets: 1. **Domain-Agnostic Semantic Ontology:** The CSIR schema, spatial operator ontology (`nearest`, `within_radius`, `within_admin_area`), and 6-dimensional validator are universally applicable to any urban or regional tourism topology. 2. **Zero-Code Geographic Migration:** Porting the system to another metropolitan area (e.g., Yogyakarta, Bandung, or Bali) requires solely updating the database bounding coordinates and relational POI records in MySQL 8.0. The core

CodeIgniter 4 compilation engine, `ST_Distance_Sphere` spatial queries, and post-generation grounding validator require zero structural modification. 3. **Dual Map Integration Agnosticism:** The presentation layer utilizes Leaflet.js with standard OGC GeoJSON specifications and public OSRM routing instances, allowing immediate redeployment without proprietary map licensing constraints.

5. CONCLUSION AND FUTURE WORK

This study designed, implemented, and evaluated a **Structured Semantic Control Layer for Reliable LLM-Mediated Spatial Querying in Web GIS**, demonstrated in Padang City. Expanding upon the exploratory spatial interaction lineage of *DTE Explorer* (Afnarius et al., 2026), the architecture decouples cognitive natural language interpretation from deterministic spatial computation, harnessing generative AI capabilities while guaranteeing complete factual and spatial fidelity.

Empirical evaluation across 40 standardized conversational scenarios, 21 automated unit tests (234 assertions), a 10,000-POI scalability stress test, and an empirical user usability study ($N = 30$) demonstrated that: 1. The **Canonical Spatial Intent Representation (CSIR)** schema and *Spatial Operator Ontology* achieved a **100.00%** semantic extraction accuracy. 2. The six-dimensional **SIR Validator** (enforcing the *No Intent Alteration* principle) and *Deterministic Spatial Query Compiler* translated user intent into secure parameterized SQL statements using the native `ST_Distance_Sphere` spatial function, achieving a **97.50%** spatial predicate execution precision. 3. The **Algorithmic Grounding Validator** and *Strict Grounding Contract* eliminated factual hallucinations completely (**0.00% Entity Fabrication Rate**), achieved **100.00% Grounding Fidelity**, and delivered a **100.00% Honest Rejection Rate** on out-of-scope requests. 4. The system demonstrated rapid end-to-end response times (**1,340.57 ms**), sub-millisecond database execution, sub-linear scalability up to 10,000 destinations (6.11 ms), an **82.43% reduction in cognitive task completion time** over conventional WIMP interfaces, and an “Excellent” **System Usability Scale (SUS) score of 84.25**.

Future work will focus on extending the control layer to multi-turn dialogue state tracking across dynamic re-routing, expanding query capabilities to multilingual international tourist dialogues, and deploying the framework across provincial multi-destination corridors.

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APPENDIX (SUPPLEMENTARY MATERIAL)

APPENDIX A: Full System Prompt for Spatial Intent Representation (SIR) Extraction

The following complete system prompt is injected into Layer 2 to enforce deterministic cognitive boundaries on the language model:

You are a spatial intent parser for a tourism Web GIS in Padang City, Indonesia.
Your task is to transform the user's natural-language query into a structured Spatial Intent Representation (SIR) in pure JSON.

RULES:

1. Return JSON ONLY. No markdown, no explanation, no other text.
2. Do not generate SQL.
3. Do not answer the user's question or provide recommendations.
4. Allowed Categories: "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" | "Kuliner" | null
5. Allowed Spatial Operators: "nearest" | "within_radius" | "within_admin_area" | "none"
6. Preserve spatial constraints exactly as expressed by user.
7. If user requests impossible things for Padang (e.g. ski, snow, casino), set is_out_of_scope: true.

SCHEMA:

```
{
  "intent": "spatial_recommendation" | "entity_lookup" | "general_inquiry",
  "entity": "tourism_object",
  "category": "Pantai" | "Pulau" | "Alam" | "Museum" | "Sejarah" | "Kuliner" | null,
  "target_name": string or null,
  "keyword": string or null,
  "spatial_operator": "nearest" | "within_radius" | "within_admin_area" | "none",
  "reference_type": "gps" | "city_center" | "poi" | "unknown",
  "radius": float or null,
  "distance_unit": "km",
  "admin_area": string or null,
  "is_free": boolean,
  "max_price": integer or null,
  "open_now": boolean,
  "open_24h": boolean,
  "sort": "termurah" | "termahal" | "terdekat" | "terbaik" | null,
  "is_out_of_scope": boolean
}
```

APPENDIX B: Full System Prompt for Grounded Natural Language Generation (Grounded NLG)

The following complete system prompt is injected into Layer 5 alongside the verified database fact tuples:

Kamu adalah asisten cerdas Web GIS Pariwisata Kota Padang.
Tugasmu adalah menjawab pertanyaan pengguna HANYA berdasarkan daftar data fakta terlampir.

KONTRAK GROUNDING KETAT (STRICT GROUNDING CONTRACT):

1. SEMUA FAKTA (nama tempat, harga tiket, jam buka, jarak) WAJIB 100% berasal dari data fakta JSON terlampir.
 2. DILARANG KERAS MENGARANG:
 - Dilarang menyebutkan objek wisata yang tidak ada di daftar data JSON.
 - Dilarang mengarang harga tiket atau jam operasional.
 - Dilarang menambahkan klaim deskriptif yang tidak tercantum pada data.
 3. Sebutkan nama objek wisata dengan cetak tebal (****Nama Objek****).
 4. Gunakan bahasa Indonesia yang santun, informatif, dan ringkas.
- [FALLBACK_NOTICE_IF_APPLICABLE]